

KAREEM MOHAMED YOUSEF

AI / Machine Learning Engineer | Software Developer

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

SUMMARY

Computer Science graduate (AASTMT) specializing in AI/ML with hands-on experience building computer-vision and full-stack systems — including a UAV-based search-and-rescue detection pipeline and a production-style AI learning platform. Comfortable across the stack: PyTorch/OpenCV for ML, Node.js/React for web. Completed the DeepLearning.AI ML and Deep Learning specializations. Looking to bring applied AI and software engineering skills to a team building real products.

EDUCATION

Bachelor of Computer Science *Oct 2022 – Jun 2026*

Arab Academy for Science, Technology and Maritime Transport (AASTMT) – South Valley
| GPA: 3.52 / 4.00 (Excellent)

TECHNICAL SKILLS

Languages: Python, C++, HTML/CSS, JavaScript, TypeScript

ML / DL: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, CNNs, MLPs, Transfer Learning, Feature Engineering, Model Regularization

Web / Backend: React, Next.js, Node.js, PostgreSQL, JWT/OAuth, REST APIs

Tools: Git, GitHub, AWS (EC2, RDS, S3), OpenCV, YOLO

EXPERIENCE

AI Team Core Member — Google Developer Student Clubs (GDSC), AASTMT *Oct 2024 – Jun 2025*

- Delivered ML/DL fundamentals sessions and mentored students through hands-on coding workshops.
- Collaborated with the AI team to build machine learning applications and educational content.

Web Development Team Member — Google Developer Student Clubs (GDSC), AASTMT *Dec 2022 – Jun 2023*

- Built responsive front-end interfaces (HTML, CSS, JavaScript) as part of a collaborative team on web development projects.

Member — ICPC Community, AASTMT *Nov 2022 – Mar 2023*

- Solved algorithmic and data-structure problems in C++ through competitive-programming training and contests.

FEATURED PROJECTS

Intelligent Search & Rescue Drone System

- AI-powered UAV system combining computer vision with flight control to detect and locate missing persons in remote, off-grid terrain.
- Designed SAOE-YOLO11m, a pruned, tiling-optimized YOLO11m variant for small-object aerial detection, trained/evaluated on the VisDrone dataset — **cut model parameters 74% (20.1M → 5.8M) and GFLOPs 26% vs. tiled YOLO11m while achieving 0.563 mAP@50, a +12.1% gain over the non-tiled baseline.**
- Built a desktop Ground Control Station (Qt/PySide) for mission planning, real-time UAV monitoring, and search-area/trajectory visualization.
- Implemented search-path optimization to cut redundant flight coverage and improve battery utilization.
- *Published as first-author-team research at the 2026 ICEENG International Conference (IEEE) — see Publications section.*

Tech: Python, PyTorch, YOLO11, OpenCV, Qt/PySide, VisDrone Dataset

DeepLearn — AI-Powered Learning Platform

- Full-stack platform for interactive technical education: content creation, smart knowledge management, and AI-assisted learning tools.
- Built a scalable Node.js backend handling authentication, content/comment/project management, and multi-level CRUD operations.
- Built a React frontend with theme switching rich-text editing, code-block rendering, and integrated AI features.
- Used PostgreSQL, JWT/OAuth authentication, file uploads, email notifications, and structured API error handling for production-grade reliability.

Tech: Node.js, React, PostgreSQL, JWT/OAuth | [Live site](#)

The Day — Student Resource Platform

- Contributed to a Next.js platform giving AASTMT CS students organized access to courses, materials, schedules, and past exams.
- Implemented dynamic semester/course pages with pinned semesters, fast search, course filtering, and optimized file previews with thumbnails.
- Worked in TypeScript with modular design and Next.js best practices; contributed UI polish, bug fixes, and component refactors.

Tech: Next.js, TypeScript | [GitHub](#) | [Live site](#)

ADDITIONAL PROJECTS

- **Friend Recommendation System** — ML-powered friend suggestions using feature engineering and supervised learning, with a GUI. [GitHub](#)
- **Cat vs. Dog Classification (CNN)** — CNN image classifier with data augmentation and documented evaluation. [GitHub](#)

- **E-commerce Mobile App** — Cross-platform Flutter app (Android/iOS/Web/Desktop) with cart, auth, and product listings. [GitHub](#)
- **Asterisk Portfolio Site** — Responsive personal portfolio (HTML/CSS/JS), deployed via GitHub Pages. [GitHub](#)

PUBLICATIONS

A. Mohamed, M. Dahb, A. Mostafa, **K. Ahmed**, S. Ahmed, A. Farhat, N. El-Bendary, and M. Hamad, "SAOE-YOLO11m: An Efficient Small Aerial Objects Expert Detection Model," *2026 ICEENG International Conference for Innovations in Intelligent Computing and Cybersecurity (IICC)*, IEEE. [IEEE Xplore](#)

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI ([Supervised ML](#), [Advanced Learning Algorithms](#))
- [Deep Learning Specialization](#) — DeepLearning.AI ([Neural Networks & Deep Learning](#), [Hyperparameter Tuning & Regularization](#), [Structuring ML Projects](#), [CNNs](#), [Sequence Models](#))

LANGUAGES

- Arabic (Native)
- English (Upper-Intermediate, B2)

LEADERSHIP

- President, The Innovation Society — Led technical initiatives and student activities promoting innovation and collaboration.
- Core Team Member, Google Developer Student Clubs (AI Team) — Delivered machine learning workshops and mentored students through hands-on projects.